Hikmet Burak Özcan, Ph.D.

Research Assistant

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Profile

My research lies at the intersection of number theory, harmonic analysis and combinatorics. I am most interested in problems where methods from one of these areas give new insight into another. In harmonic analysis, I study maximal operators and their regularity in discrete settings; my Ph.D. thesis (İzmir Institute of Technology, 2026) treats the regularity of discrete Hardy–Littlewood maximal functions. In number theory, I work on the uniform distribution of sequences and on prime numbers in rings and number fields, where Fourier analysis and exponential sums are natural tools. Combinatorial ideas, such as additive structure in the integers and higher-order difference operators, connect these directions, as in my work on the Green–Tao theorem and on the higher regularity of maximal functions. Since 2019, I have also worked as a teaching assistant at İzmir Institute of Technology, with a focus on clear explanations and guided problem solving.

Research Interests

Harmonic analysis and maximal operators · Regularity of maximal functions · Uniform distribution of sequences · Algebraic number theory · Mathematical foundations of machine learning and data science

Education

2020 – 2026 **Ph.D. in Mathematics**, İzmir Institute of Technology
Thesis: *On Regularity of Hardy–Littlewood Maximal Operators*
Supervisor: Asst. Prof. Faruk Temur

2018 – 2020 **M.Sc. in Mathematics**, Dokuz Eylül University
Thesis: *Prime Ideal Theorem on Number Fields*
Supervisor: Assoc. Prof. Haydar Göral

2011 – 2016 **B.Sc. in Mathematics**, İzmir University of Economics
GPA 3.57/4.00, graduated first in the class

Academic Position

May 2019 – present **Research and Teaching Assistant**, Department of Mathematics, İzmir Institute of Technology

- Research in harmonic analysis, uniform distribution and number theory.
- Teaching assistant for undergraduate courses and TA coordinator of MATH 141 (see Teaching).
- Weekly tutorials for core courses, exam preparation, proctoring and grading.
- Coordination of the department seminar series; maintenance of the department website.

Teaching

Fall 2026–2027 **MATH 141** Basic Calculus I – teaching assistant (TA) and TA coordinator
MATH 457 Methods of Mathematical Economics – teaching assistant (TA)

Spring 2025–2026	**MATH 332** Probability and Statistics – teaching assistant (TA)
2019 – 2026	Weekly tutorials and problem-solving sessions for Calculus I–III, Basic Calculus (MATH 141), Calculus for Engineering and Science (MATH 145), Analytic Geometry (MATH 111), Fundamentals of Mathematics, and Probability and Statistics

Teaching Expertise

Subjects taught. Single and multivariable calculus, analytic geometry, foundations of mathematics and proof, probability and statistics, and methods of mathematical economics.

Explaining mathematics. Seven years of teaching first- and second-year students in English. I explain ideas step by step with examples, and then move to formal definitions and proofs.

Problem solving. Weekly problem-solving sessions since 2019, built around worked examples, typical mistakes and exam-style questions.

Course coordination. TA coordinator of MATH 141 in Fall 2026–2027; part of exam preparation, proctoring and grading for core courses since 2019.

Publications and Preprints

2026	F. Temur, **H. B. Özcan**. Three-term difference relations do not detect uniform distribution. *Preprint*, 2026. [ResearchGate]
2026	**H. B. Özcan**. The spectrum of $(\xi\alpha^n)$ can be uncountable. *Preprint*, 2026. [arXiv:2609.07714]
2025	F. Temur, **H. B. Özcan**. The higher regularity of the discrete Hardy–Littlewood maximal function. *Preprint*, 2025. [arXiv:2504.13019]
2022	H. Göral, **H. B. Özcan**, D. C. Sertbaş. The Green–Tao theorem and the infinitude of primes in domains. *The American Mathematical Monthly*, 2022. [DOI]
2020	**H. B. Özcan**, S. Taşkın. Rings with few units and the infinitude of primes. *Hacettepe Journal of Mathematics and Statistics*, 2020. [DOI]
2020	H. Göral, **H. B. Özcan**. Several novel proofs of the infinitude of primes. *The Mathematics Student*, 2020. [ResearchGate]

Talks and Posters

Nov 2025	On higher regularity of discrete Hardy–Littlewood maximal function. *2nd Atlantic Conference in Nonlinear PDEs & Harmonic Analysis*, Instituto Superior Técnico, Universidade de Lisboa. Poster.
Sep 2025	The higher regularity of the Hardy–Littlewood maximal function. *İzmir Mathematics Day 7*, Ege University.
May 2025	On the regularity of the discrete Hardy–Littlewood maximal function. *I. Workshop on Applied Mathematics and Statistics*, İzmir University of Economics.
Jan 2025	Discretization approach to the weak boundedness of maximal convolution operators. Department Seminar, İzmir Institute of Technology.
Jul 2024	Hardy–Littlewood maximal function. Department Seminar, FernUniversität in Hagen, Germany.

May 2022	Hardy–Littlewood maximal function theorem. Department Seminar, İzmir Institute of Technology.
Nov 2021	A bridge between additive combinatorics and the infinitude of prime numbers. Department Seminar, İzmir Institute of Technology.
Oct 2020	Mathematicians will never stop to provide new proofs of the infinitude of primes. *İzmir Mathematics Days 3*, Dokuz Eylül University.
Dec 2019	Prime ideal theorem on number fields. Department Seminar, Dokuz Eylül University.
Sep 2019	Sayı cisimlerinde asal ideal teoremi (Prime ideal theorem on number fields). *32nd National Mathematics Symposium*, Ondokuz Mayıs University.
Sep 2019	A special case of Fermat's last theorem. *Algebra Workshop*, Manisa Celal Bayar University.
Sep 2019	Prime ideal theorem on number fields. *İzmir Mathematics Days 2*, Dokuz Eylül University.
Nov 2018	Congruent number problem. Department Seminar, Dokuz Eylül University.
Jun 2018	Krull–Schmidt–Remak–Azumaya theorem. *İzmir Mathematics Days 1*, Yaşar University.

Professional Service

- Referee for *The Mathematical Gazette*.
- Coordination of the department seminar series, İzmir Institute of Technology.
- Maintenance of the Department of Mathematics website, İzmir Institute of Technology.

Skills and Languages

Computing: LaTeX (lecture notes, exams, slides), Python and R (computation and statistics), Manim (mathematical animation), GitHub, HTML.

Calculators: TI-Nspire CX graphing calculator.

Languages: Turkish (native), English (professional working proficiency), Russian (beginner).